

## COMANCHE, TX, USA

WMO: 747355

Lat: 31.917N

Lon: 98.600W

Elev: 1388

StdP: 13.97

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53997

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |          | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                          |          |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD     |              |
| (a)<br>1      | (b)<br>22.9 | (c)<br>27.6 | (d)<br>8.6                    | (e)<br>9.0 | (f)<br>36.8 | (g)<br>11.7 | (h)<br>10.5 | (i)<br>41.0 | (j)<br>26.8           | (k)<br>52.2 | (l)<br>24.9 | (m)<br>53.6 | (n)<br>9.1               | (o)<br>0 | (p)<br>0.471 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>21.7            | (c)<br>100.0    | (d)<br>71.8 | (e)<br>98.7 | (f)<br>71.8 | (g)<br>96.6 | (h)<br>72.2 | (i)<br>77.2         | (j)<br>89.0 | (k)<br>76.3 | (l)<br>87.8 | (m)<br>75.4 | (n)<br>86.7 | (o)<br>9.0           | (p)<br>160 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.7                            | 136.9 | 79.5 | 73.4 | 130.7 | 78.7 | 72.8 | 128.1 | 78.6 | 41.6          | 89.8 | 40.7 | 87.2 | 39.9 | 86.2 | 80.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |           |                    |         | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |  |  |
|-------------------|----------|----------|----------------------------|-----------|--------------------|---------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|--|--|
|                   |          |          | Mean                       |           | Standard Deviation |         | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |  |  |
| 1%                | 2.5%     | 5%       | Min                        | Max       | Min                | Max     | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       |  |  |
| (a) 23.6          | (b) 20.3 | (c) 18.4 | (e) 16.6                   | (f) 103.7 | (g) 3.2            | (h) 2.2 | (i) 14.2                                           | (j) 105.3 | (k) 12.4   | (l) 106.6 | (m) 10.5   | (n) 107.9 | (o) 8.2    | (p) 109.5 |  |  |
|                   |          |          | DB                         | WB        | WB                 | WB      | WB                                                 | WB        | WB         | WB        | WB         | WB        | WB         | WB        |  |  |
|                   |          |          | 15.1                       | 78.7      | 3.5                | 1.5     | 12.5                                               | 79.8      | 10.5       | 80.7      | 8.5        | 81.5      | 5.9        | 82.6      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 66.6   | 47.9  | 51.5  | 59.5  | 66.3  | 72.8  | 81.0  | 83.8  | 83.8  | 77.9  | 67.1  | 57.5  | 49.1  |
|                                                                        | DBStd       | 15.18  | 10.38 | 11.41 | 9.46  | 7.98  | 6.58  | 4.22  | 4.02  | 4.53  | 5.61  | 8.93  | 9.35  | 10.01 |
|                                                                        | HDD50       | 493    | 164   | 109   | 29    | 5     | 0     | 0     | 0     | 0     | 0     | 7     | 37    | 142   |
|                                                                        | HDD65       | 2071   | 535   | 391   | 217   | 77    | 17    | 0     | 0     | 0     | 1     | 83    | 252   | 498   |
|                                                                        | CDD50       | 6553   | 99    | 152   | 325   | 493   | 707   | 929   | 1047  | 1049  | 837   | 537   | 263   | 115   |
|                                                                        | CDD65       | 2654   | 5     | 14    | 47    | 115   | 259   | 479   | 582   | 584   | 388   | 148   | 28    | 5     |
|                                                                        | CDH74       | 29443  | 54    | 141   | 422   | 1075  | 2362  | 5483  | 7202  | 7133  | 3928  | 1361  | 226   | 56    |
|                                                                        | CDH80       | 14594  | 7     | 30    | 103   | 366   | 932   | 2774  | 3978  | 3973  | 1896  | 481   | 46    | 8     |
| (14) Wind                                                              | WSAvg       | 8.6    | 9.1   | 9.8   | 10.4  | 10.4  | 9.1   | 8.6   | 7.6   | 7.0   | 6.6   | 8.0   | 8.6   | 8.7   |
| (15) Precipitation                                                     | PrecAvg     | 33.5   | 1.6   | 2.9   | 2.3   | 2.9   | 3.7   | 3.9   | 2.3   | 2.8   | 2.8   | 3.7   | 2.2   | 1.9   |
|                                                                        | PrecMax     | 48.1   | 3.7   | 7.5   | 4.4   | 10.2  | 11.1  | 7.5   | 6.7   | 11.7  | 5.2   | 7.1   | 7.1   | 8.3   |
|                                                                        | PrecMin     | 21.0   | 0.2   | 0.1   | 0.0   | 0.0   | 0.9   | 0.5   | 0.0   | 0.0   | 0.2   | 0.8   | 0.0   | 0.1   |
|                                                                        | PrecStd     | 7.0    | 0.9   | 1.9   | 1.3   | 2.4   | 2.2   | 2.1   | 1.9   | 3.1   | 1.7   | 2.2   | 2.1   | 1.9   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 80.4  | 83.9  | 87.5  | 94.5  | 95.4  | 100.5 | 103.1 | 102.6 | 99.5  | 93.7  | 85.5  | 80.3  |
|                                                                        |             | MCWB   | 56.6  | 61.4  | 62.5  | 64.9  | 71.9  | 71.8  | 71.3  | 73.0  | 70.3  | 70.0  | 66.3  | 59.5  |
|                                                                        | 2%          | DB     | 73.5  | 78.6  | 82.5  | 88.4  | 91.4  | 97.9  | 99.8  | 100.1 | 95.8  | 89.0  | 80.7  | 73.4  |
|                                                                        |             | MCWB   | 56.5  | 59.3  | 62.2  | 64.9  | 71.3  | 71.7  | 71.5  | 72.4  | 71.9  | 70.2  | 62.9  | 56.8  |
|                                                                        | 5%          | DB     | 69.6  | 73.3  | 79.1  | 83.9  | 89.2  | 95.1  | 97.9  | 98.7  | 93.0  | 85.6  | 75.4  | 69.8  |
|                                                                        |             | MCWB   | 55.3  | 57.8  | 61.3  | 64.6  | 70.0  | 72.4  | 72.3  | 72.3  | 71.6  | 68.5  | 61.2  | 57.1  |
|                                                                        | 10%         | DB     | 64.1  | 69.6  | 74.6  | 80.7  | 85.6  | 92.2  | 95.4  | 96.5  | 90.4  | 81.8  | 72.2  | 64.5  |
|                                                                        |             | MCWB   | 52.7  | 56.6  | 60.0  | 64.7  | 69.6  | 72.4  | 72.7  | 72.4  | 71.4  | 65.7  | 60.3  | 54.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 65.9  | 67.2  | 68.8  | 72.7  | 76.7  | 78.6  | 78.3  | 78.0  | 76.9  | 74.8  | 70.6  | 66.5  |
|                                                                        |             | MCDB   | 70.4  | 73.8  | 78.4  | 81.2  | 86.1  | 89.8  | 90.9  | 89.6  | 87.5  | 83.0  | 77.9  | 71.2  |
|                                                                        | 2%          | WB     | 61.3  | 65.0  | 67.0  | 70.8  | 75.0  | 76.9  | 76.9  | 76.8  | 75.6  | 73.2  | 67.9  | 63.5  |
|                                                                        |             | MCDB   | 67.5  | 71.7  | 75.2  | 79.6  | 84.2  | 87.7  | 89.0  | 89.6  | 85.9  | 81.4  | 73.2  | 68.4  |
|                                                                        | 5%          | WB     | 57.5  | 62.4  | 65.5  | 69.1  | 73.6  | 75.6  | 75.9  | 75.9  | 74.8  | 71.8  | 65.7  | 60.0  |
|                                                                        |             | MCDB   | 65.6  | 68.7  | 73.2  | 78.1  | 82.7  | 85.9  | 87.9  | 88.5  | 84.7  | 79.7  | 70.7  | 65.1  |
|                                                                        | 10%         | WB     | 53.7  | 59.4  | 63.7  | 67.6  | 72.4  | 74.7  | 75.0  | 75.0  | 73.8  | 70.2  | 63.6  | 56.7  |
|                                                                        |             | MCDB   | 63.5  | 66.4  | 70.5  | 75.6  | 81.1  | 85.2  | 86.9  | 87.5  | 83.2  | 78.0  | 69.3  | 63.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 21.2  | 22.0  | 22.3  | 22.5  | 20.3  | 20.9  | 21.7  | 22.1  | 21.0  | 22.4  | 22.0  | 21.0  |
|                                                                        |             | MCDBR  | 26.7  | 27.1  | 26.6  | 27.1  | 25.1  | 24.2  | 24.1  | 24.7  | 24.7  | 24.4  | 24.7  | 26.8  |
|                                                                        |             | MCWBR  | 14.4  | 14.7  | 12.1  | 12.3  | 9.1   | 6.6   | 6.3   | 6.1   | 7.0   | 9.9   | 12.6  | 14.8  |
|                                                                        |             | MCDBR  | 23.5  | 23.7  | 20.0  | 20.5  | 20.2  | 19.5  | 20.6  | 21.2  | 20.0  | 20.8  | 19.9  | 21.5  |
|                                                                        | 5% WB       | MCWBR  | 16.3  | 15.9  | 11.5  | 12.1  | 9.1   | 7.2   | 6.6   | 6.5   | 6.8   | 10.8  | 14.1  | 15.7  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.301 | 0.310 | 0.336 | 0.375 | 0.412 | 0.413 | 0.423 | 0.415 | 0.387 | 0.352 | 0.322 | 0.305 |
|                                                                        | taud        |        | 2.521 | 2.505 | 2.446 | 2.360 | 2.318 | 2.352 | 2.370 | 2.375 | 2.396 | 2.482 | 2.496 | 2.517 |
|                                                                        | Ebn at Noon |        | 295   | 302   | 300   | 290   | 279   | 277   | 273   | 275   | 279   | 283   | 284   | 287   |
|                                                                        | Edh at Noon |        | 27    | 30    | 35    | 39    | 41    | 40    | 39    | 38    | 36    | 31    | 28    | 26    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 984   | 1182  | 1526  | 1844  | 1936  | 2177  | 2181  | 1993  | 1674  | 1333  | 1034  | 871   |
|                                                                        | RadStd      |        | 105   | 173   | 123   | 159   | 192   | 178   | 155   | 135   | 138   | 201   | 133   | 114   |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (46 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +194  | N/A   |

Nomenclature: See separate page